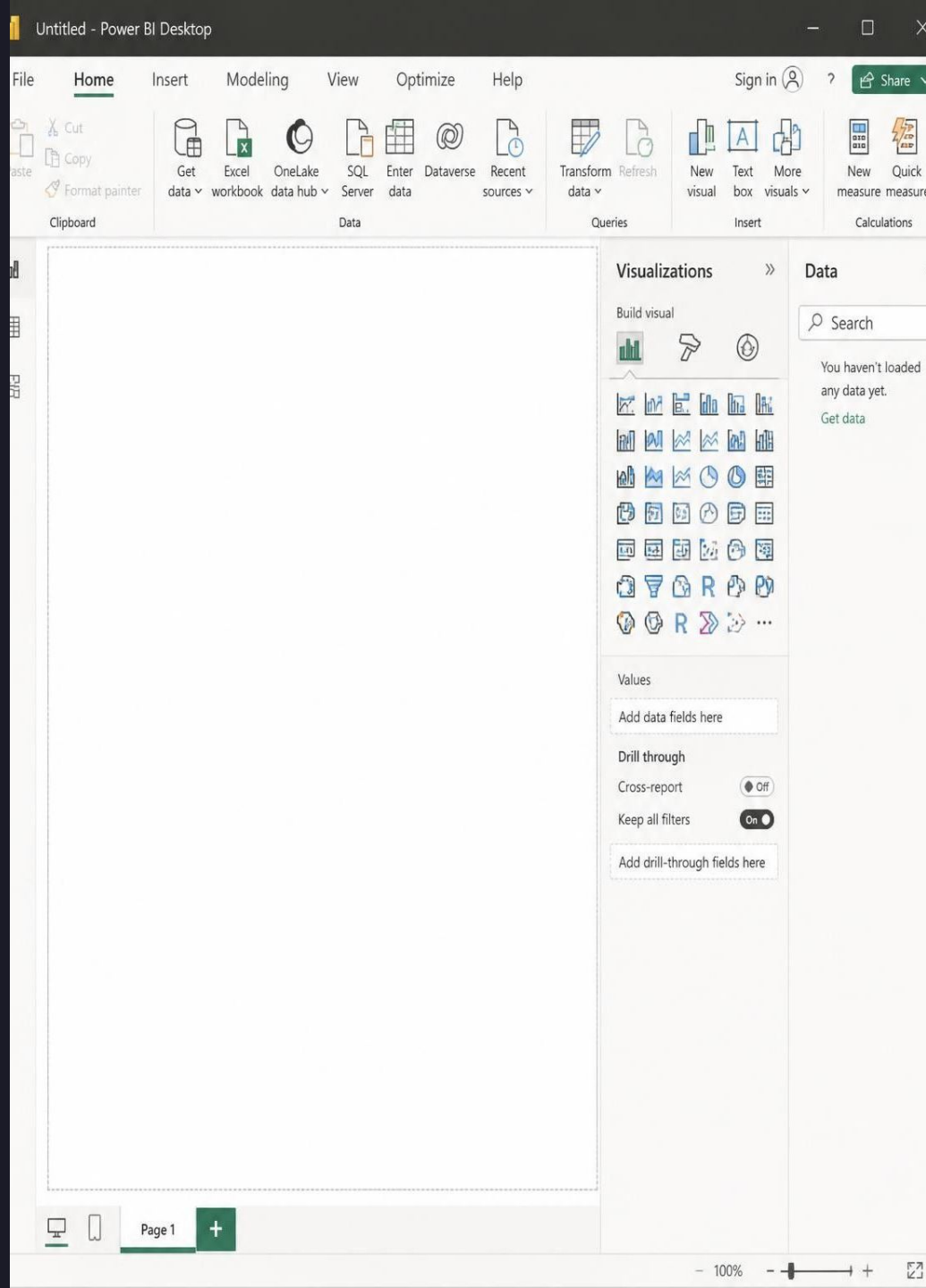
Power BI Fundamentals

From Raw Data to Business Insights



Power BI turns disconnected data into interactive insight

01

Connect

Excel, CSV, SQL, CRM, cloud platforms

02

Prepare

Clean and transform with Power Query

03

Model

Relationships and business metrics

04

Share

Interactive reports, dashboards; publish securely

Sales director scenario: revenue, customers, opportunities, products, regions and monthly performance in one place.

Why organizations use Power BI

Unified Data

Combine information from multiple business systems

Visual Analytics

Turn business records into meaningful charts

Data Refresh

Keep reports updated from source systems

Collaboration

Share governed insights across teams

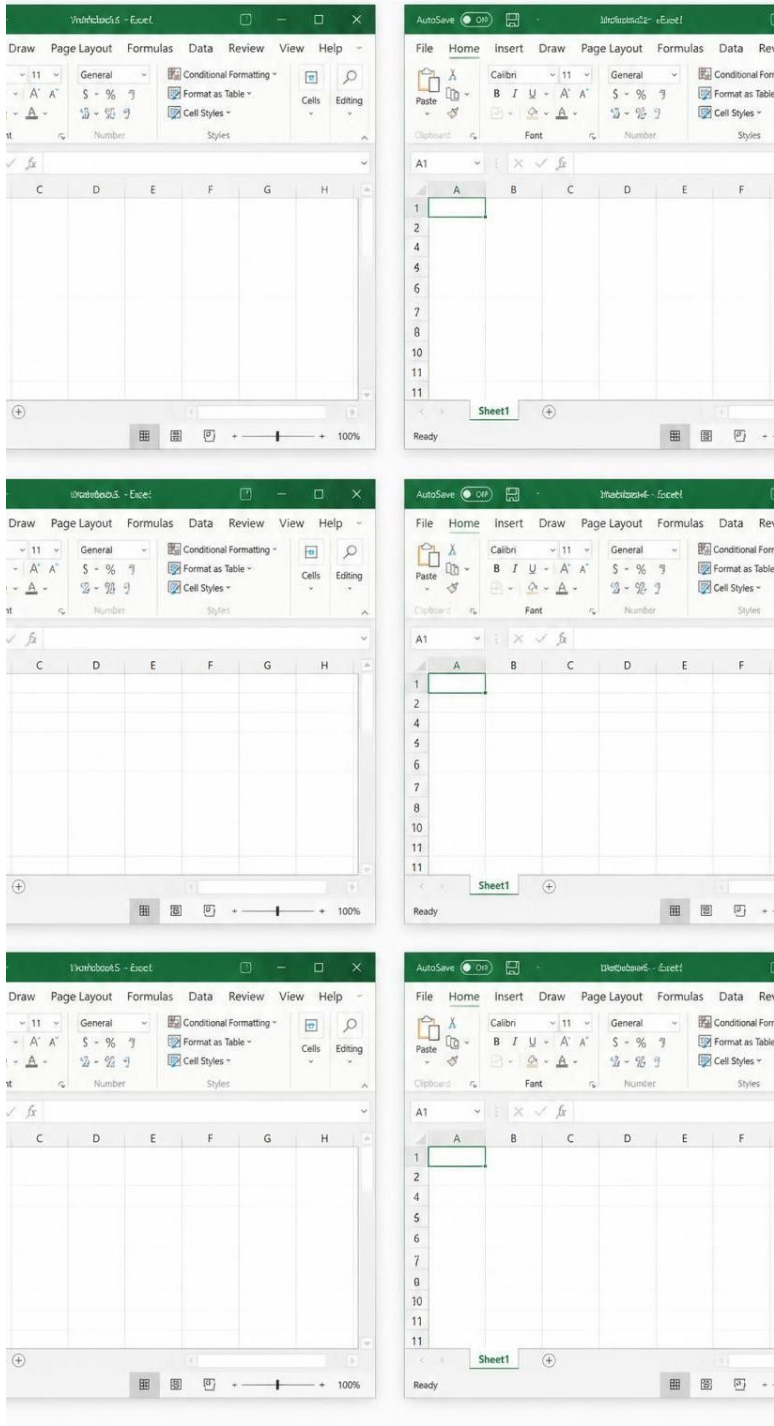
Governance

Control access and protect business data

Decision Support

Track KPIs and spot trends

One report replaces manually combining six Excel sales reports every month.



The Power BI ecosystem: from sources to decision makers

01

Sources

Excel, CSV, SQL, Dynamics 365, Azure, Fabric

02

Desktop

Connect, transform, model, report

03

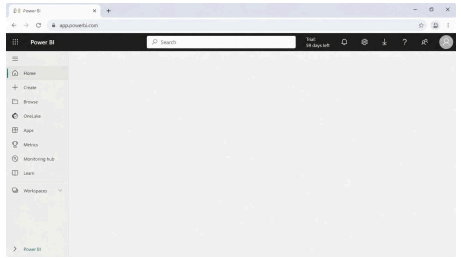
Service

Publish, share, refresh, govern

04

Business users

Web browser, mobile, Microsoft Teams, Fabric workspace



Desktop authors and models; Service supports cloud collaboration, refresh, dashboards and distribution.

Desktop authors, Service shares

Capability	Power BI Desktop	Power BI Service
Environment	Windows application	Cloud-based web application
Data connection	Yes	Yes, depending on experience
Power Query	Full authoring	Supported in applicable cloud experiences
Data modeling	Advanced	Web modeling capabilities
Report creation	Yes	Yes
Dashboards	No	Yes
Sharing	Publish to Service	Manage and share
Scheduled refresh	Not managed locally	Yes

Analyst builds the CRM Sales Report in Desktop → publishes to a Fabric workspace → sales manager opens it in Service.

Power BI is part of Microsoft Fabric

Fabric workloads

Data Factory · Data Engineering · Data Warehouse · Data Science · Real-Time Intelligence · Power BI

All on OneLake

Fabric workspaces

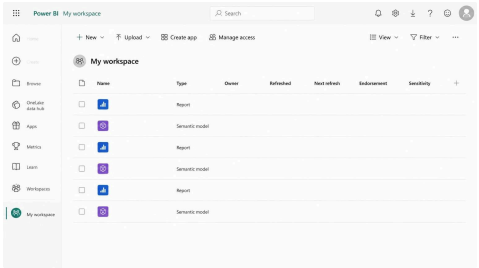
Reports · semantic models · lakehouses · warehouses · other items

Direct Lake lets supported semantic models read OneLake data efficiently.

Desktop can be used without buying Fabric capacity; cloud sharing and licensing depend on the deployment.

Reports, Dashboards and Semantic Models

Three building blocks: semantic model, report, dashboard



Semantic model

Structured business data, relationships, measures, calculation logic

Customers, products, sales

Report

Interactive, potentially multi-page analysis from a semantic model

Revenue by region

Dashboard

Single-page canvas of tiles in Service; can combine multiple reports and models

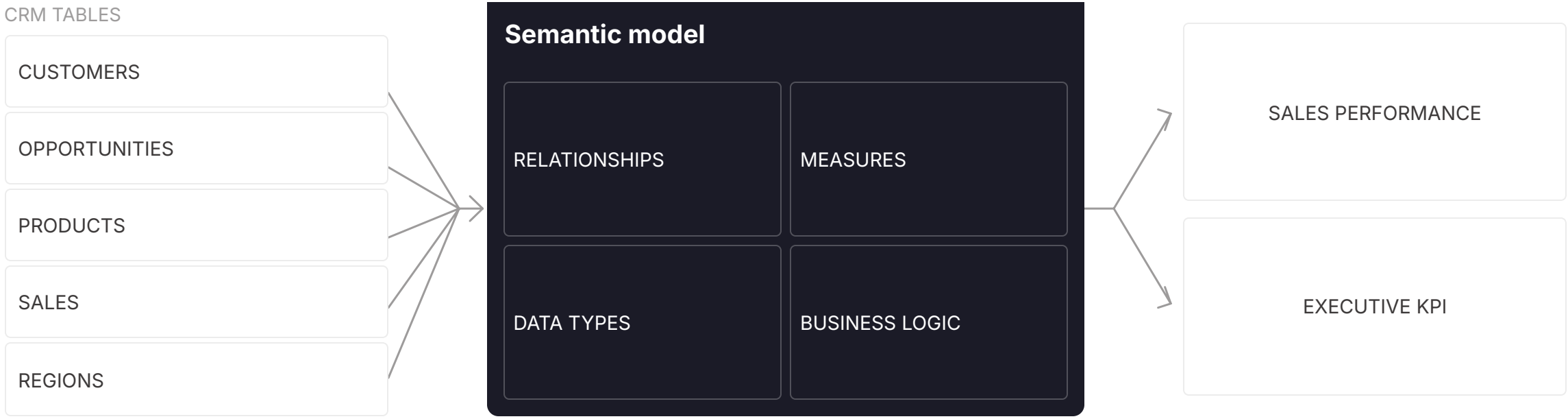
Executive KPIs

Reports dig in, dashboards keep watch

Feature	Report	Dashboard
Pages	One or more	Single page
Creation	Desktop or Service	Power BI Service
Data	One semantic model per report	Tiles from multiple sources
Interactivity	Slicers, drill-down, cross-filtering	Tile interactions and navigation
Purpose	Detailed analysis	At-a-glance monitoring

Regional manager investigates falling North sales with slicers; CEO monitors revenue, opportunities and customer growth.

A semantic model is the business-ready layer between data and reports



Total Revenue = sum of sales amounts
Total Customers = distinct customers

Won Opportunities = count with Won status
Win Rate = won divided by closed

Four storage and connectivity modes

Mode	How it works	Typical scenario
Import	Data is loaded into the semantic model	Excel/CSV sales analytics
DirectQuery	Queries are sent to the underlying source	Large operational databases
Direct Lake	Reads supported Fabric data from OneLake	Fabric Lakehouse/Warehouse analytics
Live connection	Connects to an existing semantic model	Reusing enterprise models

This module uses Import with Excel or CSV: simplest for learning Power Query, relationships and reporting.

Connecting to Data Sources

Common enterprise data sources

Excel / CSV

Local business datasets

SQL databases

SQL Server, PostgreSQL

Azure services

Azure SQL, storage, analytics

CRM / ERP

Dynamics 365, Salesforce

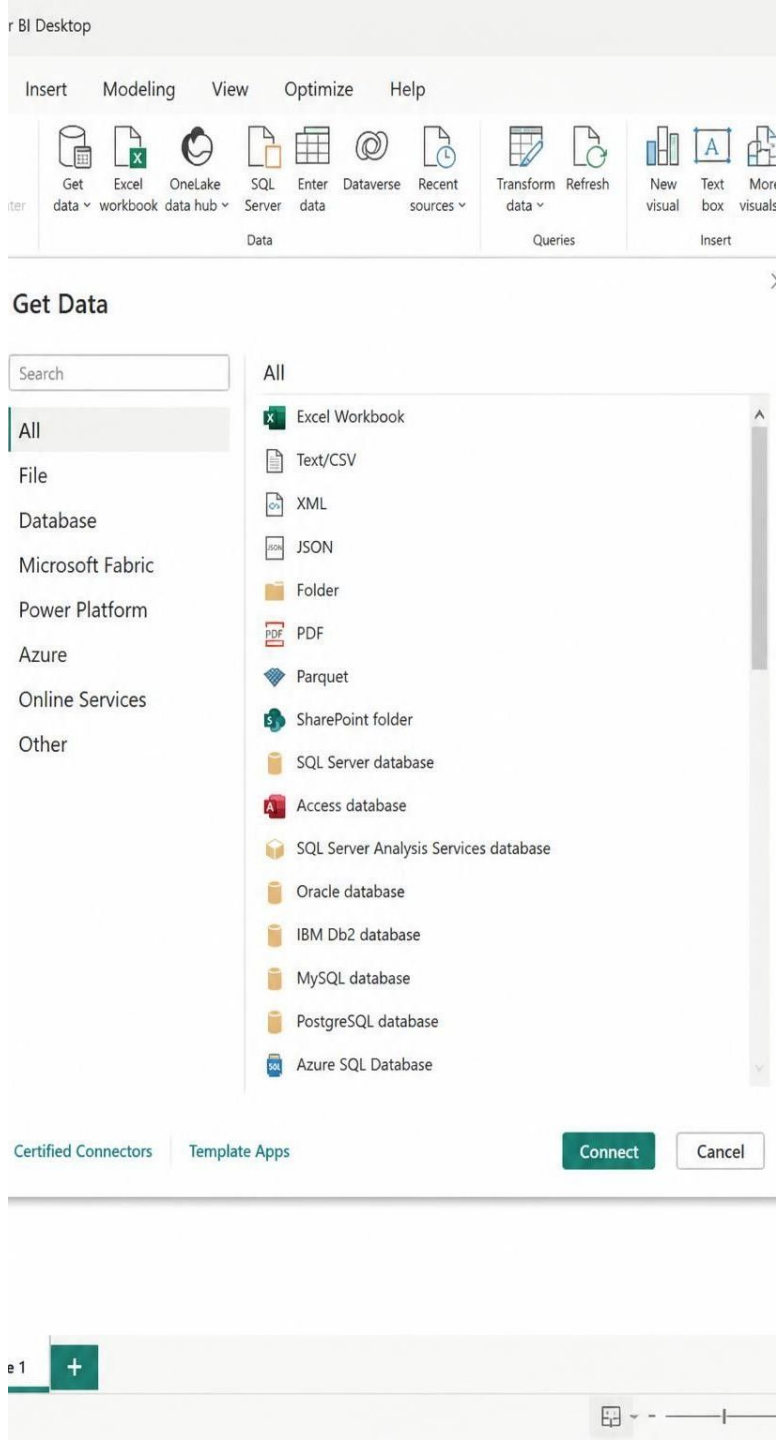
SharePoint

Files, document libraries

Microsoft Fabric

Lakehouse, Warehouse

CRM use case: Dynamics 365 customer master + SQL Server transactions + Excel monthly targets.



Connecting to Excel and CSV

1

Excel

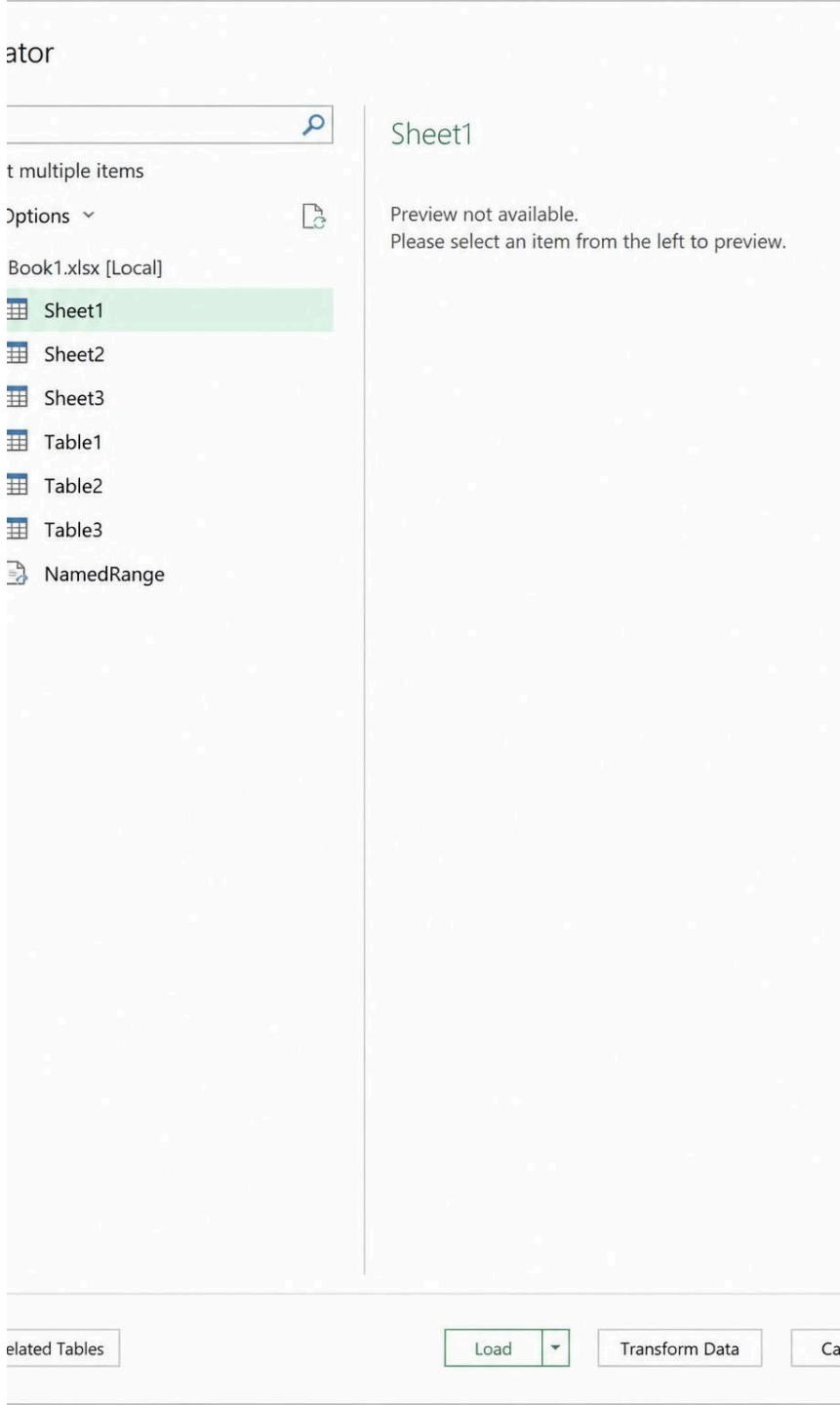
Home → Excel workbook → CRM workbook → Navigator sheets → Transform Data or Load

2

CSV

Home → Text/CSV → file → check delimiter, encoding, columns → load or transform

Verify column names, numeric formats, dates and missing values before visuals.



Power Query Fundamentals

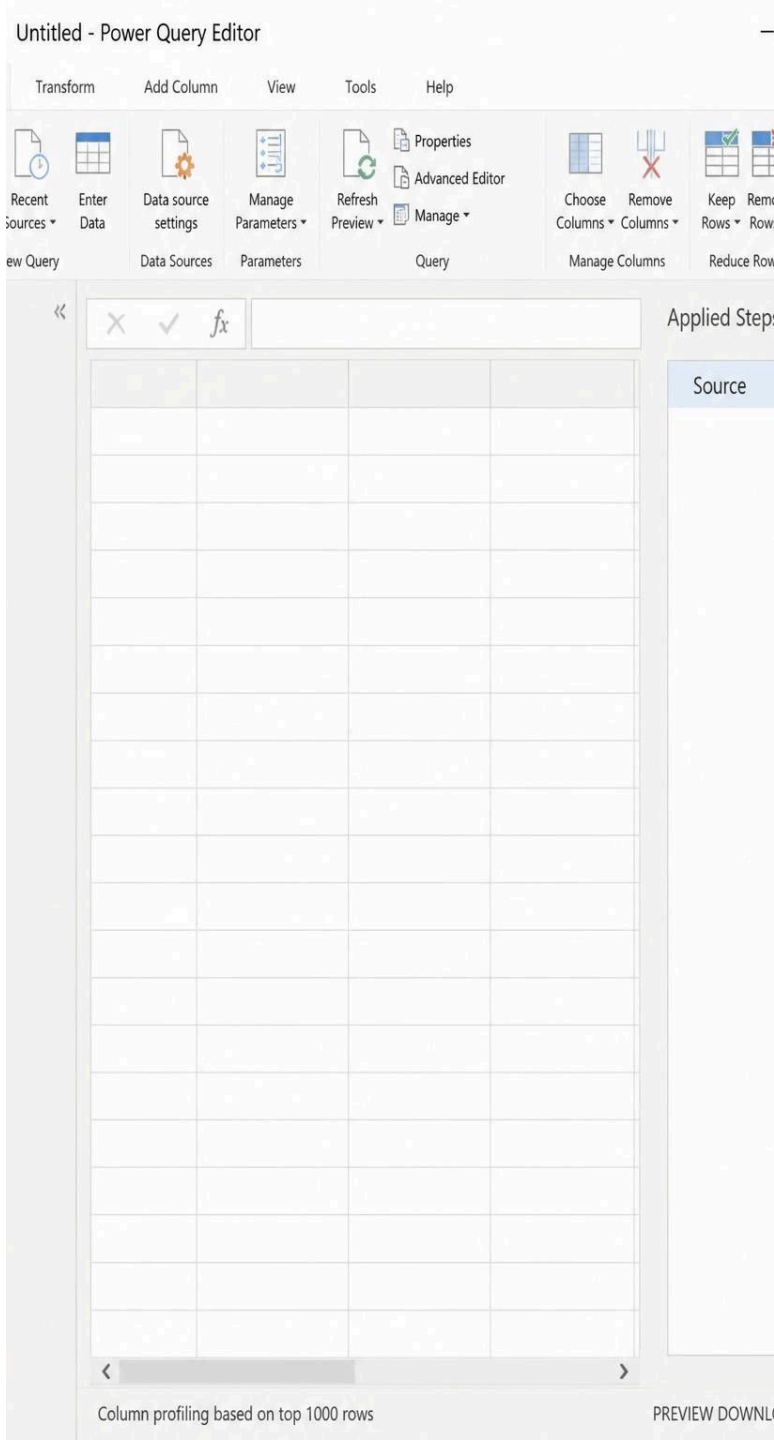
Power Query prepares data before it loads

- Remove columns and rows; rename columns; correct types
- Filter records and handle nulls
- Split, merge, replace; merge or append queries

North · NORTH · north

North

Every step is saved and repeatable.



The transformation pipeline

01

Connect

CRM sales and customer files

02

Profile

Nulls, errors, unique values, types

03

Clean

Duplicates, region names

04

Transform

Dates, currency, identifiers

05

Load

Apply to semantic model

Steps are saved and reapplied on every refresh.

Essential Power Query operations

Operation	CRM example
Remove duplicates	Eliminate duplicate customer master records
Replace values	Convert NORTH to North
Change data type	Convert CloseDate to Date
Remove columns	Remove unused export metadata
Filter rows	Exclude invalid test transactions
Merge queries	Combine sales with customer attributes
Append queries	Combine monthly sales files

Don't delete duplicate transactions without checking whether they are legitimate.

Power Query prepares, DAX calculates

BEFORE LOAD

Power Query

Clean, change types, standardize values, merge or append.

Standardize product categories

INSIDE THE MODEL

DAX

Measures, revenue, percentages, filter context and time intelligence.

Total revenue for the selected region

Fields, Data Types and Relationships

Fields and data types carry business meaning

Field	Data type	Business meaning
CustomerID	Text	Customer identifier
CustomerName	Text	Customer dimension
OpportunityID	Text	Opportunity identifier
Revenue	Decimal / Currency	Sales measure
CloseDate	Date	Time-based analysis
Region	Text	Geographic dimension
ProductCategory	Text	Product grouping
Probability	Decimal / Percentage	Opportunity likelihood

A numeric identifier is not a quantity: don't sum CustomerID.

Dimensions say how we analyze, measures say what we calculate

Dimensions

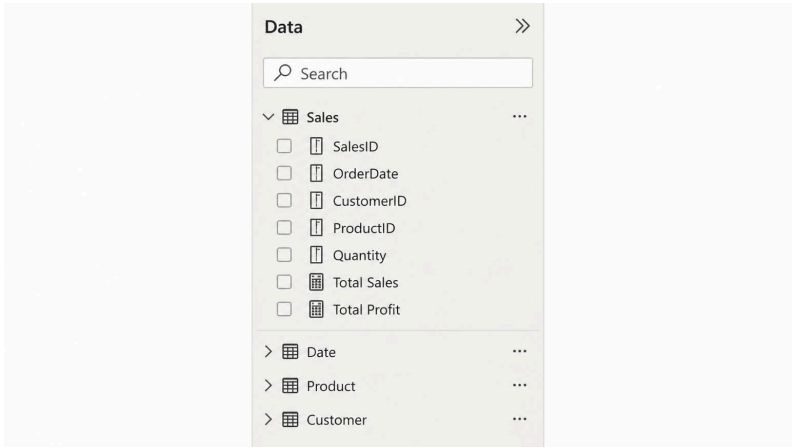
Region, Product, Customer, Salesperson, Month

Measures

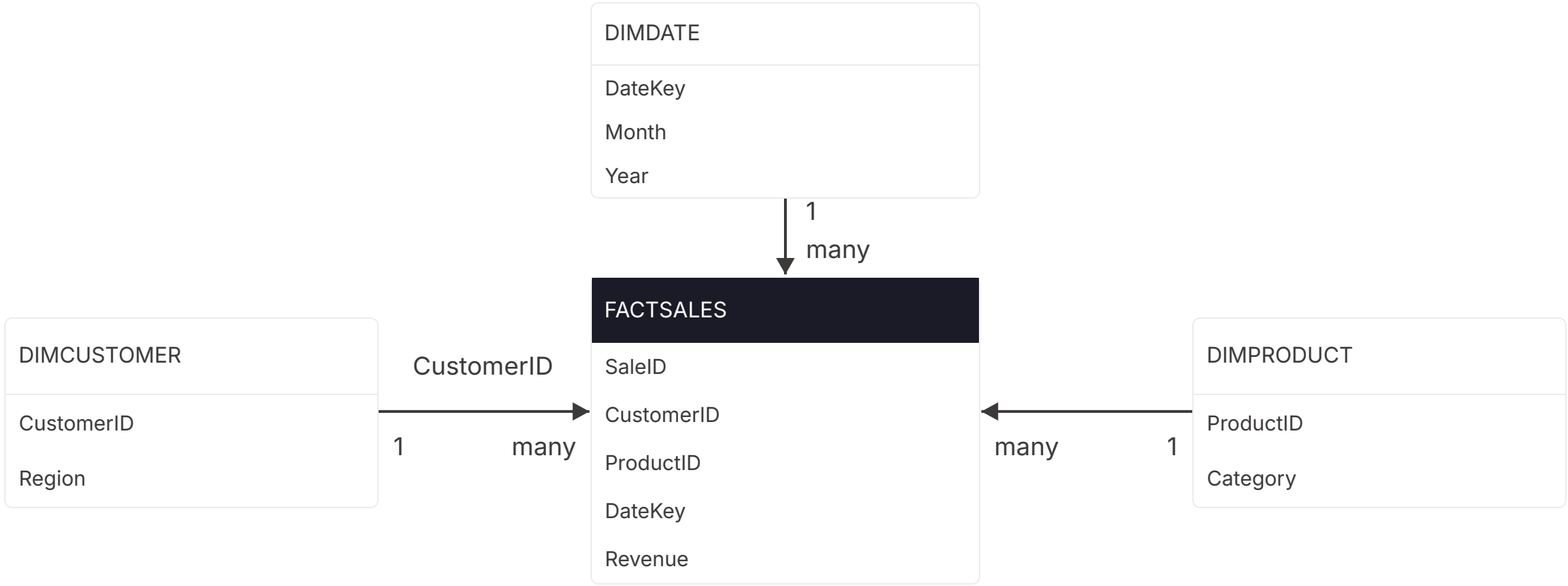
Revenue, Customer Count, Opportunity Count, Win Rate

Total revenue by region and month?

Measure: Revenue
Dimensions: Region and Month



Star schema: facts in the middle, dimensions around them



One customer, many sales. Transactions belong in facts; descriptive attributes belong in dimensions.

Cardinality and filter direction

One-to-many

One customer, many transactions

One-to-one

One record matches one record

Many-to-many

Matches on both sides; careful design

Filter direction

How filters propagate between tables

Active / inactive

Which relationship is used automatically

Best practice: one-to-many, single direction, dimensions → facts.

Visuals, Filters and Slicers

Choosing the right visual

Illustrative visual types. No sample data supplied for column, line or donut.



Bar / column

Compare revenue by region or product



Line

Revenue trends over time



Donut

Opportunity stages: Won, Open, Lost

KPI card · illustrative

₹1.24 Cr

Revenue · +12.5% vs previous period

Filters scope at three levels; slicers let users choose

Visual-level

One visual

Page-level

Applicable visuals on a page

Report-level

Applicable visuals across pages

Slicers select region, product, date or salesperson.

Illustrative demo
All Regions

₹124L

Revenue

545

Customers

Changing the region updates both.

Interactive report capabilities

Cross-filtering

Select a bar; filter related visuals

Cross-highlighting

Highlight related data elsewhere

Drill-down

Year → Quarter → Month

Drill-through

Customer or product detail page

Tooltips

Extra information on hover

Bookmarks

Saved views and navigation states

Select West: monthly revenue, product mix and customer performance.

Basic Report Development Workflow

The seven-step development lifecycle

01

Connect

CRM Excel/CSV

02

Transform

Clean and standardize

03

Model

Relationships, measures

04

Visualize

Cards, charts, tables, slicers

05

Validate

Reconcile source totals

06

Publish

Power BI Service

07

Share and refresh

Access and refresh

An attractive report with incorrect metrics is not a reliable business solution.

DAX measures: the formulas behind the KPIs

DAX (Data Analysis Expressions): formula language for analytical calculations.

```
Total Revenue = SUM(FactSales[Revenue])
Total Customers = DISTINCTCOUNT(DimCustomer[CustomerID])
Total Opportunities = DISTINCTCOUNT(FactOpportunities[OpportunityID])
Won Opportunities = CALCULATE([Total Opportunities], FactOpportunities[Stage] = "Won")
Win Rate = DIVIDE([Won Opportunities], CALCULATE([Total Opportunities],
FactOpportunities[Stage] IN {"Won", "Lost"}))
```

Assumes one identifier per opportunity; Won/Lost are closed stages. Format Win Rate as a percentage.

Publishing and Sharing

Publishing to Power BI Service

1

Save .pbix; sign in to Desktop

2

Home → Publish → destination workspace

3

Wait for publish confirmation

4

Open Service; locate report and semantic model

5

Verify published visuals and data

Publishing normally creates or updates the report and semantic model in the workspace.

Workspaces, sharing and refresh

Workspaces

Organize reports, semantic models and other Fabric items

Admin, Member, Contributor, Viewer

Sharing

Controlled access or Power BI apps

May require Pro, PPU or qualifying capacity

Refresh

Imported models stay in sync

Valid credentials; some local sources need a data gateway

Publishing alone doesn't guarantee refresh or sharing will work.

Hands-on 2:
CRM Sales Dashboard

Lab 02: build your first CRM sales dashboard

120–150 MIN

Build an interactive report to monitor performance and investigate trends.

Dataset	Fields
Customers	CustomerID, CustomerName, Region
Products	ProductID, ProductName, Category
Sales	SaleID, CustomerID, ProductID, SaleDate, Revenue
Opportunities	OpportunityID, CustomerID, Stage, EstimatedValue, CloseDate

Import → clean in Power Query → verify types → relationships → DAX → visuals → validate → publish if allowed

Deliverable: `CRM-Sales-Dashboard.pbix`

Target dashboard: CRM Sales Performance

Illustrative design · FY 2026

All Regions · **All Products**

Total Revenue

₹1.24 Cr

+12.5%

Customers

545

+8.2%

Opportunities

320

+6.4%

Monthly revenue trend

Illustrative chart slot. Monthly values are not supplied.

